2. CHARACTERISTICS OF HIGHWAY USERS

To construct an economical highway with safely flowing traffic, geometrical and physical standards have to be provided. Thus, the characteristics of human and vehicle using the highway have to be known.

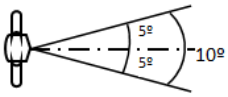
2.1 Characteristics of the driver and the pedestrian

A highway and the traffic that runs on it have three basic components: Human, Vehicle, and Road. Humans are the components that control vehicles and roads. The characteristics and behaviors of humans should be considered from the point of view of highway and traffic engineering.

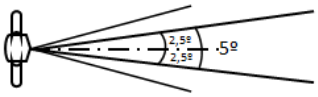
2.1.1 Normal physical characteristics

Visual characteristics: They are characteristics of the vision, one of the five senses. Characteristics like visual angle, visibility range, colorblindness, and dazzle are important for the movement of drivers and pedestrians on highways and the operation of traffic control systems. These characteristics are taken into account when the traffic signals are adjusted. Naturally, the visually disabled are legally forbidden from driving, but they are not forbidden from using the highway network as pedestrians.

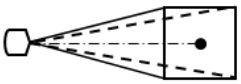
The angle on which a person can see clearly when holding the head high and looking forward is named as the **visual angle**. When the head and the eyes are remained constant, visual angle is 5º on each side of the vertical plane, making a total of 10º. There's also a horizontal plane and together they form a visual cone. **Clear visual angle** can scan each side with 2,5º.



Visual angle



Clear visual angle


Visual cone

The clear visual angle and the visual cone have an important role on arrangement and placement of traffic signals. Drivers can see the road and the other objects on the road within the visual cone. To shift the clear visual area, eyes are rotated without moving the head. The width, which can be seen with a visual angle of 10º' at a distance of 25 m, is around 4,5 m.

A person can see a wider area than the normal visual angle when looking left and right without moving the head. This angle of 180º, 90º on each side, is named the **peripheral visual angle**. Details and colors are not totally noticed under the peripheral visual angle. An area of 180º is scanned on 0,2 seconds. This is named the **angular vision speed**. The upper limit of the peripheral visual angle should not be taken into account in traffic arrangements. 65-90º are better to be chosen.

The distance at which a person is able to see clearly in a forward direction is named the **visibility distance**. This distance is important on distance and speed estimations. Under visual

impairment situations, glasses can be used to adjust visibility distance. Weather conditions also have an effect on the visibility distance.

Some people are unable to recognize colors. Color selection and contrast on the letters and the symbols of the traffic signs are determined accordingly. Black-white and yellow-black have a good contrast. Colorblindness which prevents recognizing red and green on traffic signals is a serious problem and those people are legally forbidden from driving.

Short-term visual ability loss under sudden and intense light is named **dazzle**. Dazzle is especially encountered when driving at night and at tunnel entrances/exits. To return to a normal state, at least 6 seconds are needed when entering a dark environment from light, or 3 seconds on the contrary. The intensity of the dazzle is even more in advancing (elder) years.

Distance estimation characteristic

It is important for a driver to estimate the distance of a vehicle in front or behind, and also the distance of a pedestrian crossing the street. Many traffic accidents occur because of distance estimation errors. The age, experience and education of the driver is important on distance estimation.

Auditory characteristics

Drivers and pedestrians should have at least a normal auditory level to react to the warnings and announcements made vocally on highways. Driving license is not given to people who have less then required auditory level. However, people who use some devices to increase their hearing ability to a desired level can also obtain a driving license.

Perception-reaction characteristics

The total time needed for a driver or a pedestrian to see a danger on the highway, percieve it, decide what to do and apply his/her reaction is named as the perception-reaction time. Perception-reaction time is important for determining vehicle stopping (braking) distances, calculating phases on signalized intersections, determining the place of traffic signs.

Perception-reaction time varies for people. Also perception-reaction time of a person can vary under different conditions and times (sleepy, drunk, cheerful, disprited, morning, night, etc.). As the person gets older or tired, this time increases. On laboratory exams, average value of the perception-reaction time is calculated as 0.5-2.5 seconds. Various values are used worldwide while in Turkey, 0.75-1 seconds are used. However AASHTO proposes an average value of 2.5 seconds.

2.1.2 Temporary physical characteristics

Characteristics that a person is not always under the effect of, are called as the temporary physical characteristics. The effects of these characteristics can be seen on a short or long term and are results of fatigue, disease and usage of pleasure-inducing substances.

Fatigue directly affects the perception-reaction time and the ability to maneuver. A fatigued person can unwillingly make mistaken movements. For this reason, long distance drivers are legally forbidden to drive longer than specifed times. Also, being mentally fatigued may cause more serious results than being physically fatigued.

Pleasure-inducing substances like alcoholic beverages or drugs decrease the judgment and movement ability of drivers severely. The same situation is also valid for pedestrians. It is legally forbidden to drive under the effects of pleasure-inducing substances, and penalties are very high.

Diseased people are generally calmed (still) more than normal and their movement abilities decrease. As a result of the disease or effect of the medicine taken, perception-reaction times and manoeuvre abilities of drivers and pedestrians may significantly decrease.

2.1.3 Mental characteristics

Some of the mental characteristics are congenital while some others are learnt. They can be summarized as intelligence, knowledge, good manners, experience, reading and foreign language knowledge.

Intelligence is a natural and a hereditary characteristic. It is effective on perceiving traffic signs and reacting well on encountered situations. Knowlegde and good manner is obtained later on. They can be increased with reading, studying and observing. Experience is gained by time. Experienced drivers give better decisions and get less involved in accidents than unexperinced drivers.

A person has to be literate to understand notifications and warning signs on highways. Illiterate people can not obtain a driving license. Foreign language knowledge is not so important as most of the traffic signs are usually the same worldwide. However, foreign language knowledge provides convenience for drivers on international highways.

2.1.4 Psychological characteristics

It is not easy to know the psychological characteristics of people exactly under their conditions. However some various psychological characteristics affect the behaviour of the user of the highway. Some of them can be listed as, gathering attention, corresponding with the rules, being patient and calm, tendency to high speed and accidents.

Gathering attention is required for safe driving. Distracted people have more of a possibility of getting involved in an accident. Fatigue and occupation also divert attention. Some people have a habit of breaking the rules. This situation is natural-born for some people, while being uneducated is the reason for others. Two other important characteristics for drivers and pedestrians are being patient and calm. Especially in traffic jams and accidents, precipitation, and uncontrolled behavior make the situation even worse. These characteristics are usually hereditary. However, they can be improved with education.

While some people seem to be normal in terms of physical and mental characteristics, they have a tendency to speed up and be involved in accidents. While this behaviour is accepted normal on high-speed highways, under normal traffic conditions, it is an indication of anormality.

2.1.5 Some determinations about the driver characteristics

- Drivers working in a suburban area have a tendency to drive faster than those working in an urban area.

- Long distance drivers, new automobile drivers, drivers between the age of 25-29 violate the speed limits more than others.
- Ages between 40-49 for men and 30-39 for women are proper for normal velocities.
- If the traffic is flowing slowly, drivers are affected more by their speeds.
- Faster drivers have a tendency to follow the center line, it gets difficult to follow the lane lines.
- Men and women drive at similar speeds.
- Drivers have a tendency to drive slower if they have a relative as a passenger in the vehicle.
- People who drive faster than 75 km/h are more involved in accidents than people who drive around 45-65 km/h.
- Drivers approaching a city drive faster than those who are leaving a city.

2.1.6 Pedestrian Walking Speed

Pedestrian-vehicle accidents may occur on traffic especially when pedestrians are crossing a street. For this reason it is important to know the walking speed of pedestrians. It is especially used on determining the green light time for pedestrians of traffic lights.

Some factors like age, sex, traffic conditions affect the pedestrian velocities. Also the physical and the psychological characteristics of the pedestrians affect the walking speed. Some values are given on the table below.

Age of the Pedestrian	Pedestrian walking speed (m/sec)			
	Under heavy traffic conditions		Under light traffic conditions	
	Normal	Fast	Normal	Fast
0-7	1,13	2,40	1,29	2,33
7-15	1,34	2,40	1,16	2,91
15-25	1,34	2,15	1,24	2,33
>25, male	2,40	2,40	2,33	2,33
>25, female	1,20	2,15	1,16	2,33

2.2 Vehicle characteristics

Another component of the highway and its traffic is the vehicles. Highway vehicles can be categorized into two: motorized vehicles and unmotorized vehicles. The characteristics of various types and properties of vehicles are mainly mechanical engineers' tasks. Here, only some vehicle characteristics of the highway will be considered.

2.2.1 Motorized Vehicle Types

According to Highway Traffic Law 2918, cars that carry men, animals, or freight on highways are named vehicles.

Vehicles running using a motor is named as a motorized vehicles, while vehicles running using men or animal power are named as ummotorized vehicle. The term "car" on the law is the general name for motorized, unmotorized, special purpose vehicles, construction equipment and rubber tyred tractors running on highways.

Motorized vehicles are; motorcycles, automobiles, minibuses, buses, vans, trucks, tractors, rubber tyred tractors, construction equipments, land vehicles, special purpose vehicles (fire

truck, ambulance, funeral, radio, cinema, television, library, research cars and motorized vehicles which pull or carry, lift broken down or damaged vehicles).

Construction equipment are equipment used for constructions or similar agricultural, industrial, public works, national defence organizations and they are not used to carry human, animal or freight on highways. Tractors are motorized vehicles that pull trailers or semi trailers.

Another classification can be made as light vehicles and heavy vehicles. Automobiles, minibusses and vans are classified as light vehicles while busses and trucks are classified as heavy vehicles. Heavy vehicles usually have more than 2 axles and manufactured to carry more than 3,5 tons. The definitions, properties and carriage limits of motorized vehicles are stated on Highway Traffic Law and regulations.

Design vehicle

The design vehicle, which is determined to provide the accomodation of dominant vehicle type on the highway, is one of the design criteria to take into account the vehicle sizes, weights, operational properties and traffic density.

The most important vehicle characteristics that affect the highway design are the minimum rotation paths of the front-outer and the inner-rear tyres and as well as the axle distance.

Vehicles are one of the main components of highway safety, and their characteristics and performances have an important effect on determining highway design criteria.

Metric

Design Vehicle Type	Symbol	Dimensions (m)											
		Overall			Overhang								Typical Kingpin to Center of Rear Axle
		Height	Width	Length	Front	Rear	WB_1	WB_2	S	T	WB_3	WB_4	
Passenger Car	P	1.3	2.1	5.8	0.9	1.5	3.4	–	–	–	–	–	–
Single Unit Truck	SU	3.4-4.1	2.4	9.2	1.2	1.8	6.1	–	–	–	–	–	–
Buses													
Inter-city Bus (Motor Coaches)	BUS-12	3.7	2.6	12.2	1.8	1.9[a]	7.3	1.1	–	–	–	–	–
	BUS-14	3.7	2.6	13.7	1.8	2.6[a]	8.1	1.2	–	–	–	–	–
City Transit Bus	CITY-BUS	3.2	2.6	12.2	2.1	2.4	7.6	–	–	–	–	–	–
Conventional School Bus (65 pass.)	S-BUS 11	3.2	2.4	10.9	0.8	3.7	6.5	–	–	–	–	–	–
Large School Bus (84 pass.)	S-BUS 12	3.2	2.4	12.2	2.1	4.0	6.1	–	–	–	–	–	–
Articulated Bus	A-BUS	3.4	2.6	18.3	2.6	3.1	6.7	5.9	1.9[a]	4.0[a]	–	–	–
Trucks													
Intermediate Semitrailer	WB-12	4.1	2.4	13.9	0.9	0.8[a]	3.8	8.4	–	–	–	–	8.4
Intermediate Semitrailer	WB-15	4.1	2.6	16.8	0.9	0.6[a]	4.5	10.8	–	–	–	–	11.4
Interstate Semitrailer	WB-19*	4.1	2.6	20.9	1.2	0.8[a]	6.6	12.3	–	–	–	–	13.0
Interstate Semitrailer	WB-20**	4.1	2.6	22.4	1.2	1.4-0.8[a]	6.6	13.2-13.8	–	–	–	–	13.9-14.5
"Double-Bottom"-Semitrailer/Trailer	WB-20D	4.1	2.6	22.4	0.7	0.9	3.4	7.0	0.9[a]	2.1[a]	7.0	–	7.0
Triple-Semitrailer/ Trailers	WB-30T	4.1	2.6	32.0	0.7	0.9	3.4	6.9	0.9[c]	2.1[c]	7.0	7.0	7.0
Turnpike Double-Semitrailer/Trailer	WB-33D*	4.1	2.6	34.8	0.7	0.8[a]	4.4	12.2	0.8[a]	3.1[a]	13.6	–	13.0
Recreational Vehicles													
Motor Home	MH	3.7	2.4	9.2	1.2	1.8	6.1	–	–	–	–	–	–
Car and Camper Trailer	P/T	3.1	2.4	14.8	0.9	3.1	3.4	–	1.5	5.8	–	–	–
Car and Boat Trailer	P/B	–	2.4	12.8	0.9	2.4	3.4	–	1.5	4.6	–	–	–
Motor Home and Boat Trailer	MH/B	3.7	2.4	16.2	1.2	2.4	6.1	–	1.8	4.6	–	–	–
Farm Tractor[i]	TR	3.1	2.4-3.1	4.9[g]	–	–	3.1	2.7	0.9	2.0	–	–	–

Note: Since vehicles are manufactured in U.S. Customary dimensions and to provide only one physical size for each design vehicle, the values shown in the design vehicle drawings have been soft converted from numbers listed in feet, and then the numbers in this table have been rounded to the nearest tenth of a meter.

* = Design vehicle with 14.63 m trailer as adopted in 1982 Surface Transportation Assistance Act (STAA).
** = Design vehicle with 16.16 m trailer as grandfathered in with 1982 Surface Transportation Assistance Act (STAA).
a = Combined dimension is 5.91 m and articulating section is 1.22 m wide.
b = Combined dimension is typically 3.05 m.
c = Combined dimension is typically 3.05 m.
d = Combined dimension is typically 3.81 m.
e = This is overhang from the back axle of the tandem axle assembly.
f = Dimensions are for a 150-200 hp tractor excluding any wagon length.
g = To obtain the total length of tractor and one wagon, add 5.64 m to tractor length. Wagon length is measured from front of drawbar to rear of wagon, and drawbar is 1.98 m long.

- WB_1, WB_2, and WB_4 are the effective vehicle wheelbases, or distances between axle groups, starting at the front and working towards the back of each unit.
- S is the distance from the rear effective axle to the hitch point or point of articulation.
- T is the distance from the hitch point or point of articulation measured back to the center of the next axle or center of tandem axle assembly.

Exhibit 2-1. Design Vehicle Dimensions

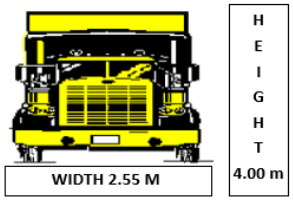
US Customary

Design Vehicle Type	Symbol	Dimensions (ft)											
		Overall			Overhang								Typical Kingpin to Center of Rear Axle
		Height	Width	Length	Front	Rear	WB_1	WB_2	S	T	WB_3	WB_4	
Passenger Car	P	4.25	7	19	3	5	11	–	–	–	–	–	–
Single Unit Truck	SU	11-13.5	8.0	30	4	6	20	–	–	–	–	–	–
Buses													
Inter-city Bus (Motor Coaches)	BUS-40	12.0	8.5	40	6	6.3[a]	24	3.7	–	–	–	–	–
	BUS-45	12.0	8.5	45	6	8.5[a]	26.5	4.0	–	–	–	–	–
City Transit Bus	CITY-BUS	10.5	8.5	40	7	8	25	–	–	–	–	–	–
Conventional School Bus (65 pass.)	S-BUS 36	10.5	8.0	35.8	2.5	12	21.3	–	–	–	–	–	–
Large School Bus (84 pass.)	S-BUS 40	10.5	8.0	40	7	13	20	–	–	–	–	–	–
Articulated Bus	A-BUS	11.0	8.5	60	8.6	10	22.0	19.4	6.2[a]	13.2[a]	–	–	–
Trucks													
Intermediate Semitrailer	WB-40	13.5	8.0	45.5	3	2.5[a]	12.5	27.5	–	–	–	–	27.5
Intermediate Semitrailer	WB-50	13.5	8.5	55	3	2[a]	14.6	35.4	–	–	–	–	37.5
Interstate Semitrailer	WB-62*	13.5	8.5	68.5	4	2.5[a]	21.6	40.4	–	–	–	–	42.5
Interstate Semitrailer	WB-65** or WB-67	13.5	8.5	73.5	4	4.5-2.5[a]	21.6	43.4-45.4	–	–	–	–	45.5-47.5
"Double-Bottom"-Semitrailer/Trailer	WB-67D	13.5	8.5	73.3	2.33	3	11.0	23.0	3.0[a]	7.0[a]	23.0	–	23.0
Triple-Semitrailer/ Trailers	WB-100T	13.5	8.5	104.8	2.33	3	11.0	22.5	3.0[c]	7.0[c]	23.0	23.0	23.0
Turnpike Double-Semitrailer/Trailer	WB-109D*	13.5	8.5	114	2.33	2.5[c]	14.3	39.9	2.5[c]	10.0[c]	44.5	–	42.5
Recreational Vehicles													
Motor Home	MH	12	8	30	4	6	20	–	–	–	–	–	–
Car and Camper Trailer	P/T	10	8	48.7	3	10	11	–	5	19	–	–	–
Car and Boat Trailer	P/B	–	8	42	3	8	11	–	5	15	–	–	–
Motor Home and Boat Trailer	MH/B	12	8	53	4	8	20	–	6	15	–	–	–
Farm Tractor[f]	TR	10	8-10	16[g]	–	–	10	9	3	6.5	–	–	–

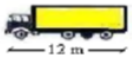
* = Design vehicle with 48 ft trailer as adopted in 1982 Surface Transportation Assistance Act (STAA).
** = Design vehicle with 53 ft trailer as grandfathered in with 1982 Surface Transportation Assistance Act (STAA).
[a] = Combined dimension is 19.4 ft and articulating section is 4 ft wide.
[b] = Combined dimension is typically 10.0 ft.
[c] = Combined dimension is typically 10.0 ft.
[d] = Combined dimension is typically 12.5 ft.
[e] = This is overhang from the back axle of the tandem axle assembly.
[f] = Dimensions are for a 150-200 hp tractor excluding any wagon length.
[g] = To obtain the total length of tractor and one wagon, add 18.5 ft to tractor length. Wagon length is measured from front of drawbar to rear of wagon, and drawbar is 6.5 ft long.

- WB_1, WB_2, and WB_4 are the effective vehicle wheelbases, or distances between axle groups, starting at the front and working towards the back of each unit.
- S is the distance from the rear effective axle to the hitch point or point of articulation.
- T is the distance from the hitch point or point of articulation measured back to the center of the next axle or center of tandem axle assembly.

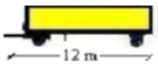
Exhibit 2-1. Design Vehicle Dimensions (Continued)



Refrigerated structure width in refrigerated vehicles 2.60 m



Motor vehicles other than bus: 12.00 m



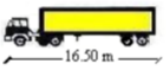
Trailers: 12.00 m



Buses with two axles: 13.50 m



Buses with more than two axles: 15.00 m



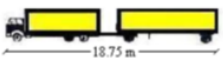
Semi-trailer vehicles: 16.50 m



Articulated buses: 18.75 m



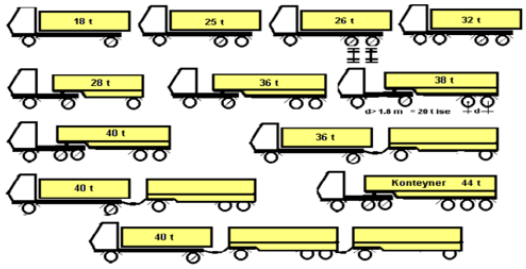
Buses with trailer: 18.75 m



Trucks with trailer: 18.75 m

Allowed limits of total weights for vehicles

Vehicle types	Weight (ton)
Motor vehicles and trailers with two axels	18.00
Motor vehicles with three axels (*Under condition in Type Approval Regulation)	25.00 (26.00*)
Semi-trailer three-axel vehicles and articulated buses	28.00
Motor vehicles with four axels	32.00
Four-axel vehicles with trailer and semi-trailer	36.00
Four-axel vehicles with semi-trailer, the weight of semi-trailer axel group is 20 tons	38.00
Five- or more than five-axel trains with trailer and semi-trailer	40.00
As an activity of combined transport, 40-ft size ISO container carrying motor vehicles having three axels and two- or three-axel semi-trailer	44.00

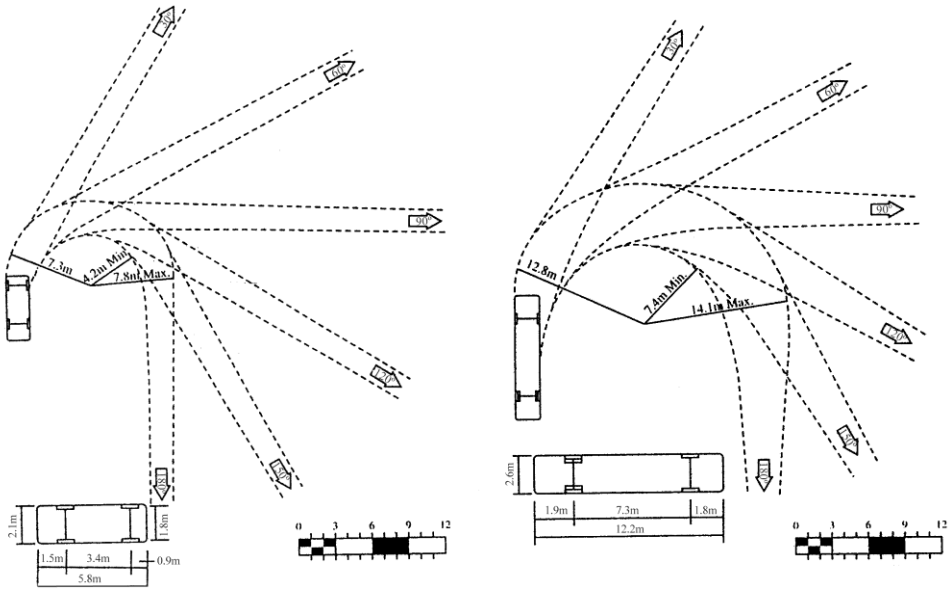


2.2.3 Minimum vehicle rotation radius

The minimum rotation radius on constant movement of vehicles are various. The minimum safe rotation radius varies according to the type, size and the rotation speed of the vehicle. Rotation radius are especially important in designing urban intersections.

Minimum design rotation radius is mostly used for very slow vehicles (usually with a speed less than 15 km/h). For higher speeds, dynamical effects are taken into account so that minimum rotation radius is neglected. However, minimum design rotation radius can be used on u-turn and parking areas.

The rotation radius is around 7 m for small vehicles, 13 m for trucks, and 15 m for semi-tractors for 90º turns. As the most unfavorable situation should be considered in the project/design phase, turns with high radii are not taken into account. For turns with higher velocities, detailed calculations have to be made by taking parameters like project speed, superelevation, and lateral friction coefficient into account. The rotation characteristics of vehicles are given as a template, as in the figure below, and these rotation radii are considered in the project phase.



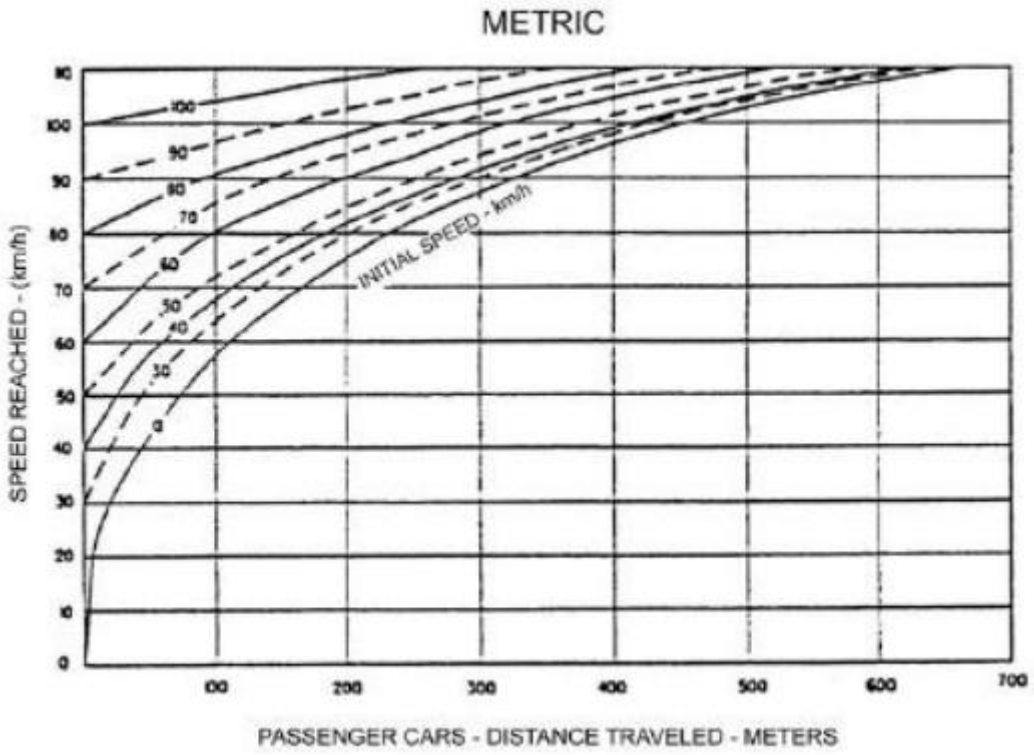
2.2.4 Acceleration and deceleration of vehicles

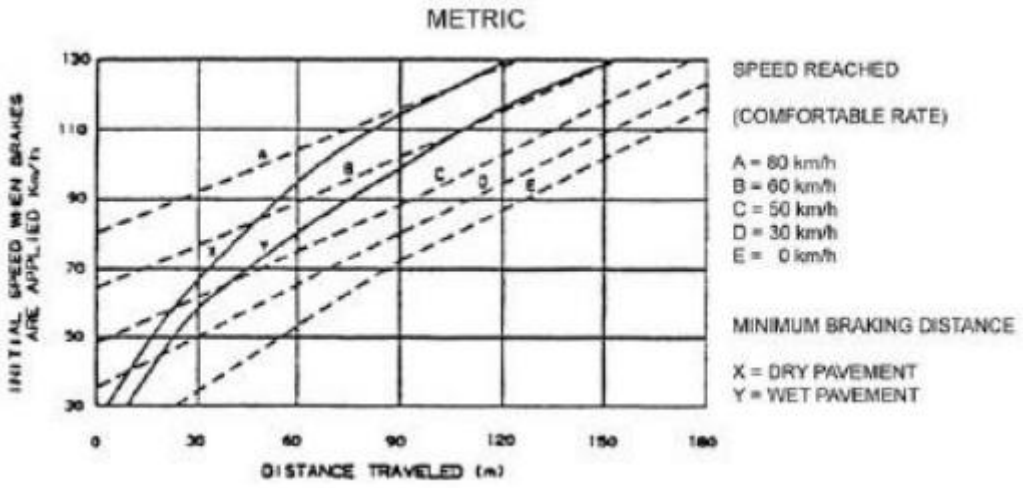
Acceleration characteristics of vehicles are important for deciding the distance needed to overtake a vehicle, the time needed to pass through an intersection, and projecting acceleration/deceleration lanes.

Acceleration depends on the motor power, power/weight ratio, characteristics and internal friction of drivetrains, speed, and the grade of the highway.

Weight/power ratio has a special importance on acceleration performances of vehicles. As the weight of a vehicle having a constant power decreases, the acceleration ability increases. Acceleration is important in intersection design, calculating the minimum passing sight distance, and the length of acceleration and decceleration lanes. Acceleration values vary between 0,5-3,0 m/sec² for various types of vehicles.

Acceleration value is higher on lower speeds. For example, below 40 km/h, for automobiles they vary between 1,80-2,70 m/sec² and for trucks between 0,60-0,90 m/sec². As the accelerator pedal of a moving vehicle is released, the velocity decreases because of the resistances. The deceleration value for a vehicle moving with a velocity of 100 km/h is around 1 m/sec² for this situation.





2.2.5 Operating costs of motorized vehicles

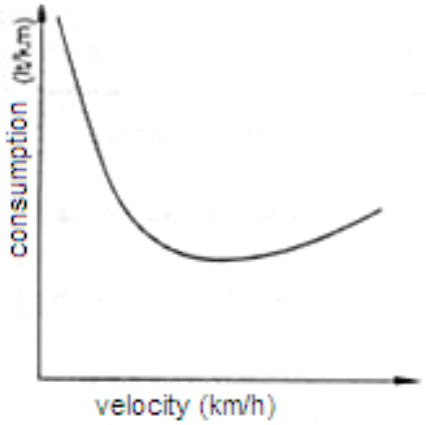
A projected highway has to be economical in several ways as well as being able to serve with determined purposes. One of the indicators of economy is the low operation costs of vehicles moving on the highway.The operation costs of vehicles can be categorized as fixed and variable expense items. Variable expenses emerge as vehicle starts to move. Fixed expenses are independent of vehicle movements.

In general, variable expenses are fuel, oil, tyre, maintenance, repair and replacement part costs; while fixed expenses are taxes, insurance and rent (park, garrage, etc.).

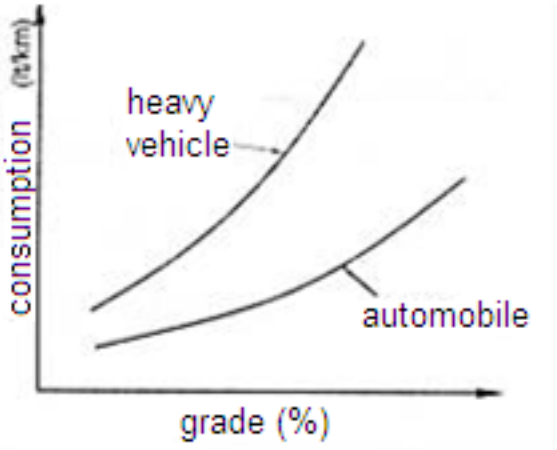
Properties of the highway does not affect the constant expenses. However, physical and geometrical properties of the highway directly affect the expenses about vehicle movements.

Parameters about the highway that affect the cost are grade, pavement type and condition, horizontal curves and the condition of the traffic. These parameters generate the resistances that prevent vehicle movements. Resistances directly affect the vehicle speed.

There's an optimum speed range for vehicles that minimizes the fuel consumption. Comsumption increases over or below this speed range. For example, for automobiles, 60-80 km/h (2000-2500 cycle/min) range is the ideal speed range for minimizing the fuel consumption.



Fuel consumption increases with the grade of the highway. The effect of the grade is even higher for heavy vehicles. However, on descending sections of a highway heavy vehicles consume lesser amount of fuel but the difference is not as much as expected because vehicles have to apply brake for safe travel.



To reflect the grade's effect on a vehicle's operating cost, graded length is converted into equivalent straight length, as seen in the table below. For example, for a highway section of 2 km with a %5 grade, operating costs of trucks are approximately $L_{eq,\ truck}$ = 2.(2,25)= 4,5 km of straight highway. The operating costs of light vehicles in the same section are approximately,

$L_{eq,\ light\ vehicle}$ =2.(1,14)= 2,28 km of straight highway.

Vehicle type	Grades (%)										
	0,00-0,50	0,60-1,50	1,60-2,50	2,60-3,50	3,60-4,50	4,60-5,50	5,60-6,50	6,60-7,50	7,60-8,50	8,60-9,50	9,60-10,50
Light vehicle	1,00	1,02	1,04	1,06	1,09	1,14	1,18	1,24	1,32	1,40	1,49
Heavy vehicle	1,00	1,25	1,50	1,75	2,00	2,25	2,50	2,75	3,00	3,25	3,50

The type of the pavement is effective on both fuel consumption and the tyre wear. Fuel consumption is higher on non-rigid pavement. Increasing speed wears out tyres faster. Also as the smoothness of the pavement increases, maintenance and repair costs decrease. For this reason concrete and asphalt pavements are ideal.

Vehicles consume more fuel on curves than on alignments because of decreased velocity. The motor efficiency loss on deceleration is effective on this consumption. As the curve radius decreases, speed is limited, and fuel consumption increases at lower speeds. This effect of the horizontal curves can be reflected in the fuel consumption by converting the curve length into the equivalent straight length. For this reason, a distance of L_{ek} (m) must be added to the curve length. The excess length can be calculated with the empirical formula below:

$$L_{ek} = 0{,}077 \quad (V_p^2 - V_k^2) + \left(\frac{L}{V_p} - \frac{V_k}{V_p}\right) \quad (V_p \quad V_k)$$

L_{ek}: The excess length to be added to the curve length [m]

V_p: The speed of the vehicle on a straight section [km/h]. Project speed can be used.

V_k: The speed of the vehicle on the curve [km/h]. Constrained speed can be used.

D: The curve length [m]

The excess length of a curve on a highway section, having the curve angle of 60º, accepted maximum superelevetion of 8% and the project speed of 90 km/h is calculated and shown on the table below using the empirical formula.

V_p=90 km/h, q_{max}=8%, Δ= 60 °										
R (m)	50	75	100	150	200	250	300	350	400	449
$V_ç$ (km/h)	30,0	36,8	42,5	52,0	60,1	67,2	73,6	79,5	85,0	90,0
L (m)	52,4	78,5	104,7	157,1	209,4	261,8	314,2	366,5	418,9	470,2
L_{ek} (m)	638,6	611,2	579,3	507,7	429,8	347,9	263,1	176,1	87,4	0,0

On congested traffic conditions, fuel consumption increases. For this reason operating costs on urban roads are higher than suburban roads. Another important factor on operating costs is driving with proper tehniques which decrease the fuel consumption 30-35%.

2.2.6 Nonmotorized Vehicles

Vehicles which are operated using human or animal power is named as nonmotorized vehicles. Nonmotorized vehicles, that are used rarely because of the advanced technology, are more common in rural areas. Most common type is the 2 or 4 wheeled carts. Oxcarts have almost vanished.

Nonmotorized vehicles decrease the traffic safety of highways. As their speeds are much lower than motorized vehicles, they create disequilibrium on traffic flow and result in overtaking problems and rear-end accidents.

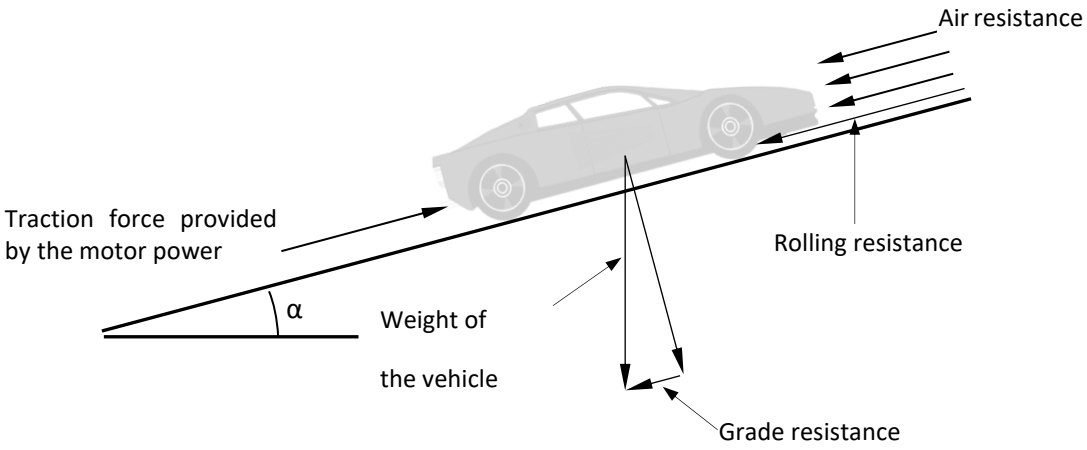
3. VEHICLE MOVEMENTS

A traction force is needed to make a vehicle move. Vehicles receive their traction force from human, animal or the motor assembled on them. On motor vehicles, motion is realised when the traction force provided from the motor power overcomes some resistances.

Resistances are the main factors, except the brake, that prevent the motion of the vehicle. Even if a vehicle is in motion as the result of the traction force overcoming the resistances, when the gas pedal is released, it stops at a time. For this reason, the driver constantly keeps his/her foot on the gas pedal. In this regard, driving can be explained as keeping the vehicle constantly in motion, overcoming the resistances under variable conditions (road geometry, rolling surface, sight distance, etc.). The most important outcomes of the driving process are safe transportation and changes in vehicle operating costs. As mentioned before, driving techniques directly affect the vehicle's operating costs.

3.1 Resistances affecting motor vehicles

The main resistances the motor vehicles encounter while starting to move or during their motion are; trundling (rolling) resistance, air resistance, grade resistance, curve resistance and acceleration resistance. These resistances, except the curve and the acceleration resistance is shown schematically on the figure below.



In highway and vehicle engineering, these resistances are tried to decrease as much as possible to decrease vehicle operating costs.

Rolling resistance (W_y)

The main reasons for the rolling resistance are;

- Deformation of the tyres on their contact points with the pavement
- Air circulation around the tyres
- Friction between the vehicle's power trains.

The major component of the rolling resistance is the deformation of the tyres.

The size of the resistance depends on;

- The situation of the pavement
- The inner pressure of the tyre
- The elasticity of the tyre
- Velocity of the vehicle

The rolling resistance can be calculated as:

$$W_y = \mu_r.Q$$

W_y : rolling resistance (N)

μ_r : rolling resistance coefficient (N/N or N/kN)

Q : Weight of the vehicle (N)

If the speed of the vehicle is below 50 km/h, rolling resistance coefficient can directly be taken from the table below.

Pavement type	μ_0 coefficient
Concrete roads and light case asphalt pavement	0,010-0,020
Asphalt Mcadam pavement	0,020-0,025
Crushed rock and gravel pavement	0,030-0,040
Durax-cube pavement	0,040-0,050
Dry and tight earth roads	0,030-0,060
Loose ground, sandy, slimy surfaces	0,015-0,060

If the speed of the vehicle is over 50 km/h, the values on the table are improved with the given formula. The rolling resistance coefficient decreases as the speed of the vehicle increases, as seen in the formula.

$$\mu_t = \mu_0 [1\text{-}0{,}01\ (V\text{-}50)]$$

Air resistance (W_h)

Vehicles move inside an air tunnel and are exposed to the air pressure effects in the tunnel. The main reasons for the air resistance are;

- The air pressure affecting perpendicular to the vehicle's cross-section in the opposite direction of the vehicle movement,
- Friction resulting from the turbulence air around the vehicle,
- The air around, below, and behind the vehicle wheels,
- The air is generated by the radiator and ventilator in the vehicle.

According to the aerodynamic laws, air resistance can be formulated as:

$$W_h = c.\rho.F.\left(\frac{V}{3.6}\right)^2$$

Here;

W_h : Air resistance (kg)

c : A coefficient related to the vehicle's shape and surface smoothness

ρ : The specific mass of the air at sea level and 15 ºC (ρ = 0,125 kg.sec²/m^4)

F : The cross-section area of the vehicle, perpendicular to the moving direction (m^2). Approximately, it can be taken as a specific percentage of the vehicle's height times width, usually, F=(0,8~0,9).b.h.

V : Speed of the vehicle (km/h)

In windy weather, the velocity is;

If the vehicle is moving in the same direction with the wind→$V=(V_{veh}-V_{wind})$

If the vehicle is moving in the opposite direction of the wind →$V=(V_{veh}+V_{wind})$

If the air resistance coefficient K is used instead of the (c. ρ) term in the formula;

$$W_a = K.F.\left(\frac{V}{3,6}\right)^2$$

Limits of K and F values for various types of vehicles are given in the table below. If the unit of the air resistance is taken as Newtons, the formula becomes:

$$W_a = 0,75.K.F.V^2$$

Vehicle type	K (kg.sec²/m⁴)	F (m²)
Racing cars	0,010-0,015	1,5-2,0
Normal automobiles	0,015-0,030	1,5-2,6
Busses	0,025-0,050	4,0-6,5
Trucks	0,050-0,070	3,0-6,0

Grade resistance (W_e)

The reason for the grade resistance is the grade of the highway. The grade resistance is the parallel component of the vehicle weight to the rolling surface. On an ascending (upgrade) grade, it makes the movement more difficult (positive), while on a descending (downgrade) grade, it makes the movement easier (negative).

$W_e = Q.\sin\alpha$

As the α angles are small on highways, $\sin\alpha \approx tg\alpha$ can be taken. Thus,

$W_e = Q.tg\alpha$

can be written.

tgα is also the grade of the road (g_r) on percent. So, for the grade resistance, it can be used.

W_e : Grade resistance (N) $$W_e = \frac{Q.g_r}{100}$$

Q : Weight of the vehicle (N)

α : The angle of the road with the horizontal plane (in degrees)

g_r : Grade of the road as percent (%)

Curve resistance (W_k)

While moving on horizontal curves, the frontal wheels are rotated with an angle to the axis. Curve resistance occurs at the contact point of the frontal wheels with the pavement surface. Curve resistance increases with the curvature and the speed of the vehicle. In other words, it has an inverse proportion to the curve radius and a direct proportion to the vehicle speed. Tables have been prepared for curve resistance based on various experiment results. The table below shows the curve resistance affecting a vehicle moving on asphalt pavement in a good situation according to different curve radii and speed values.

Curve radius (m)	Veloctiy (km/h)	Curve resistance (N)
350	80	180
350	96	360
175	48	180
175	64	540
175	80	1080

Acceleration resistance (W_i)

Acceleration resistance is the result of increasing or decreasing speed. The first principle of the dynamics is, F=(m. a) where the F is the force affecting on an object, m is the mass of the object and a is the acceleration. The F force, emerges on speed changes is named as the acceleration force and it creates a resistance which the motor has to overcome. Acceleration resistance has to be overcome while the vehilce increases its speed so it takes a positive value.

On the contrary, as the vehicle decreases its speed, the acceleration resistance helps the motor so it takes a negative value. The amount of this resistance depends on the weight of the vehicle and the acceleration.

It can be calculated as:

$$Wi = \frac{Q}{g}.a$$

W_i : Acceleration resistance (N)

Q : Weight of the vehicle (N)

g : Gravitational acceleration (g=9,81 m/sec^2)

a : Acceleration of the vehicle (m/sec^2)

If the value of the gravitational acceleration is taken as g=10 m/sec^2,

$$Wi = \frac{Q}{10}.\gamma$$

3.2 Power, traction force, and motion on motorized vehicles

To be able to move a motor vehicle, the power generated by its motor has to be transferred to the wheels using the motor trains. The tyres which provide the first motion are named motor (propelling) wheels. Rear-wheel or frontal-wheel drive shows the rear and the frontal wheels being motors, respectively. The power of the motor vehicle can be written as:

$$N_m = Z_{tr}.v$$

N_m : Motor power of the vehicle (Watt)
Z_{tr} : The traction force on motor wheels (N)
v : Speed of the vehicle (m/sec)

If the traction force is written by taking the power in units of kW (kilowatt) and the speed in km/h;

$$Z_{tr} = 3600.\frac{N_m}{V}$$ is obtained.

Actually, a specific amount of the power generated by the motor is lost while being transferred to the wheels because of the friction of some parts of the power trains.

To take this loss into account, the power generated by the motor has to be multiplied with a specific η efficiency coefficient.

η coefficient is smaller than 1 and varies between 0,80~0,90.

So the formula takes the form of:

$$Z_{tr} = 3600.\frac{N_m}{V}.\eta$$

Z_{tr} is the force rotating the wheels in order to start the vehicle's movement. As the traction force is transferred to the wheels, a friction occurs between the road surface and the wheels.

With the help of this friction and according to the action-reaction principle (effect of the traction force-reaction of the road surface) the wheel starts to rotate.

To start the movement, the resistances have to be overtaken by the traction force. This is named as the first condition of the motion and can be formulated as:

$$Z_{tr} > \Sigma W$$

$$Z_{tr} > W_y + W_h + W_e + W_k + W_i$$

If the vehicle is moving with a constant speed on a curveless and graded section, the formula can be written as:

$$Z_{tr} > W_y + W_h + W_e$$

If the vehicle moves with a constant speed on slopeless and straight sections, the formula can be written as:

$$Z_{tr} > W_y + W_h$$

Also, to start a stopping vehicle moving, the friction force between the pavement surface and the motor wheels has to be bigger than the resistances. This is named as the second condition of the movement and can be written as:

$$Q_m.f > \Sigma W$$

Q_m : The adherence weight or, in other words, the weight of the vehicle that is carried by the motor wheels (N). Usually, for heavy vehicles with rear-wheel drive, Q_m=2/3.Q; for automobiles, Q_m=1/2.Q; for vehicles with four-wheel drive, Q_m=Q.

f : Sliding friction coefficient

$Q_m.f$: Friction (adherence) force (N)

According to the two conditions of motion, for a stopping vehicle, the following situations can be written:

1. $Z_{tr} < \Sigma W < Q_m.f \rightarrow$ The vehicle does not move (no motion)

2. $\Sigma W < Z_{tr} < Q_m.f \rightarrow$ The vehicle starts to move with the normal motion (rolling condition)

3. $\Sigma W < Q_m.f < Z_{tr} \rightarrow$ The vehicle starts to move while spinning (forward motion with spinning)

4. $Q_m.f < \Sigma W < Z_{tr} \rightarrow$ The vehicle spins and does not move (spinning without motion)

A vehicle stopping on a straight road section is not affected by the air, grade, curve, acceleration resistances. So, instead of the formula above, $W_y=\mu_r.Q$, rolling resistance can also be written. However, if the vehicle is on a graded section, the grade resistance must also be included.

In the first situation, the primary principle of the movement can not be fulfilled, so the vehicle fails to move. Even if the friction force can overcome the resistances, the traction force transmitted to the motor wheels is not enough. To overcome this problem, the power of the vehicle has to be increased, or the resistances have to be decreased. If the vehicle does not move when the gas pedal has been pushed, the motor power is not transmitted to the wheels (there can be a malfunction), or the motor power fails to overcome the resistances due to overloading. Overloaded trucks failing to move is a good example of this situation.

$$\Sigma W < Z_{tr} < Q_m.f$$

Since the both principles of the movement is fulfilled in the second situation, the vehicle starts to move with its normal movement.

$$\Sigma W < Q_m.f < Z_{tr}$$

In the third situation, the principles of the movement is fulfilled but as the traction power transmitted to the motor wheels is greater than the friction force, the movement is obtained while skidding. When the traction force is suddenly transmitted to the wheels, or in other words, the gas pedal has been pushed suddenly and powerfully at low gears, traction force overcomes and this situation occurs.

$$Q_m.f < \Sigma W < Z_{tr}$$

In the fourth situation, while the first principle of the movement is fulfilled, the second is not. In other words, even if the traction force that can overcome the resistance is transmitted to the wheels, the friction force that helps the first movement is lower than the resistance. In this situation, either the motor weight has to be increased or the friction coefficient must be increased in order to increase the friction force. Practically, either the weight on the motor axles is increased, or substances like sand, gravel, or cinder are poured under the skidding wheels.

3.3 Sight distances

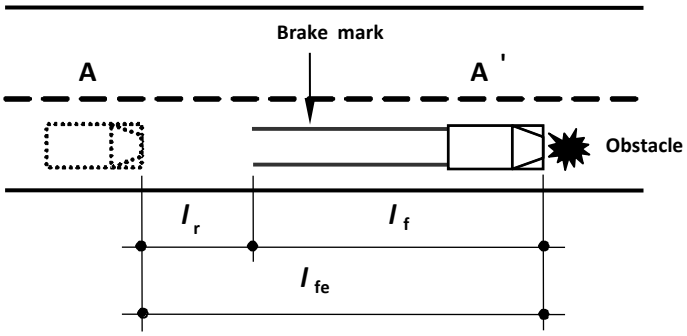
Drivers have to make some manouevres that the traffic requires. During these movements, they should not endanger their and other vehicles' safety. Movements like acceleration, decceleration, stopping, overtaking a vehicle have to be made within a safe distance. These distances are used on geometrical design of highways and determining the location of the traffic signs.

As already mentioned, the clear distance a driver can see during his drive on the highway is named the sight distance. The sight distance is determined by factors like weather conditions (rain, snow, fog) or day-night drive. Also, substructure components like crest vertical curves, overpasses, facilities on horizontal curves, vegetation, cut slopes, and various surface types also affect the sight distance. If the sight distance is limited in these places, the speed limit of overtake prohibition is applied, and drivers are warned to use the traffic signs.

On the other hand, sight distance have to be provided for drivers to take precaution against an obstacle (pedestrian, vehicle or an object) they can encounter.

3.3.1 Stop-sight distance

The minimum sight distance for a driver to avoid hitting an obstacle on his/her lane is named the stop-sight distance. This distance is also called brake safety distance or stopping distance and consists of two parts. The first one is the distance covered during perception-reaction time and named as the response distance. The other one is the brake distance covered on brake time. This distances for a braking vehicle is seen on the figure below.



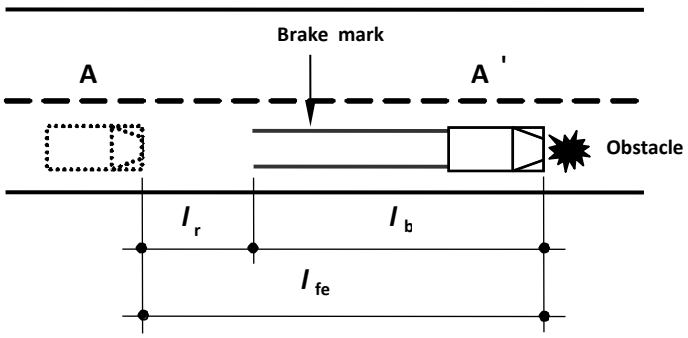
After the driver sees an obstacle on the road that can endanger his drive, he perceives it, designs the potential precautions, and decides and applies his/her decision during the perception-reaction distance. This distance is the function of the constant speed of the vehicle and the perception-reaction time. It can be formulated as:

$$l_r = v.\ t_r$$

l_r : Reaction length (m)
v : Speed of the vehicle (m/sec)
t_r : Perception-reaction time (sec)

This distance increases with the speed of the vehicle or with the perception-reaction time of the driver. If the speed is taken as km/h, the formula can be written as:

$$l_r = 0{,}278.V.\ t_r$$



The brake distance which is the second part of the stop-sight distance shows the distance covered between the driver pushes on the brake pedal and the vehicle stops. Motor wheels are

blocked when the brake is applied and a black brake mark occurs on the pavement due to the friction.

By using the preservation of energy, the kinetic energy of a moving vehicle is transformed into work by the appliance of the brake, and thermal energy comes out. So,

$$\frac{1}{2}.\frac{Q}{g}.v^2 = Q.l_f.(f \pm s)$$

can be written. The first part of the ($f \pm s$) term shows the energy coming out due to the friction, while the second part shows the energy coming out due to the grade. In this equation,

Q : Weight of the vehicle (N)
g : Gravitational acceleration (g=9,81 m/sec^2)
l_f : Brake length (m)
v : Velocity of the vehicle (m/sec)
f : Sliding friction coefficient
s : Grade of the highway.

By simplifying the equation,

$$l_f = \frac{v^2}{2.g}.\frac{1}{(f \pm s)}$$

can be written. If the velocity is taken as km/h, and the gravitational acceleration as g=9,81 m/sec^2, the formula becomes:

$$l_f = 0{,}00394.\frac{V^2}{(f \pm s)}$$

Finally, the brake safety distance for a vehicle moving with the velocity of V km/h can be written as the sum of the two distances:

$$l_{fe} = 0{,}278.V.t_r + 0{,}00394.\frac{V^2}{(f \pm s)}$$

As the grade g_r on the formula acts like a resistance on ascending (upgrade) sections it takes a positive sign and as it helps the movement on descending (downgrade) sections, it takes a negative sign.

The sliding friction coefficient (f) between the wheels and the pavement surface is determined by some factors. These are the type, and smoothness speed of the vehicle. On dry and smooth pavement, friction coefficient is also high. Some values of friction coefficient for various pavement types and road surfaces are seen in the following table.

Type of the pavement	Condition of the highway surface	Sliding friction coefficient
Asphalt	Dry	0,60
	Wet&Clean	0,35
	Wet&slightly muddy	0,25
Concrete	Dry	0,75
	Wet&Clean	0,55
	Wet&slightly muddy	0,40
Parquet	Dry	0,60
	Wet&Clean	0,40
	Wet&slightly muddy	0,30
Basalt small parquet	Dry	0,45
	Wet&Clean	0,25
	Wet&slightly muddy	0,15
Granite small parquet	Dry	0,50
	Wet&Clean	0,35
	Wet&slightly muddy	0,25
Tarmac	Dry	0,50
	Wet&Clean	0,35
	Wet&slightly muddy	0,25
Rolled basalt	Kuru	0,70
	Islak temiz	0,50
	Islak hafif çamurlu	0,40
Wooden parquet	Dry	0,55
	Wet&Clean	0,40
	Wet&slightly muddy	0,30
Snow	Dry&Chained	0,55
	Dry&Unchained	0,45
	Wet&Chained	0,20
	Wet&Unchained	0,10
Ice	Dry&Chained	0,35
	Dry&Unchained	0,15
	Wet&Chained	0,15
	Wet&Unchained	0,05

As a highway is being designed, stop-sight distances are calculated for the most inconvenient situations. The most inconvenient situation for the pavement is being wet or even frozen.

On the other hand, the stop-sight distance formula are same worldwide, however as the driver perception-reaction times vary, the stop sight distances calculated also vary. As already mentioned, t_r=0,75 sec is used in our country. So for a straight (s=0) and dry asphalt pavement (f=0,60) highway, reaction, brake and brake safety distances for various speeds are given in the following table.

V (km/h)	v (m/sec)	l_r (m)	l_b (m)	l_{bs} (m)
10	2,78	2,09	0,66	2,74
15	4,17	3,13	1,48	4,61
20	5,56	4,17	2,63	6,80
25	6,94	5,21	4,10	9,32
30	8,33	6,26	5,91	12,17
35	9,72	7,30	8,04	15,34
40	11,11	8,34	10,51	18,85
45	12,50	9,38	13,30	22,68
50	13,89	10,43	16,42	26,84
55	15,28	11,47	19,86	31,33
60	16,67	12,51	23,64	36,15
65	18,06	13,55	27,74	41,30
70	19,44	14,60	32,18	46,77
75	20,83	15,64	36,94	52,58
80	22,22	16,68	42,03	58,71
85	23,61	17,72	47,44	65,17
90	25,00	18,77	53,19	71,96
95	26,39	19,81	59,26	79,07
100	27,78	20,85	65,67	86,52
105	29,17	21,89	72,40	94,29
110	30,56	22,94	79,46	102,39
115	31,94	23,98	86,84	110,82
120	33,33	25,02	94,56	119,58
130	36,11	27,11	110,98	138,08

Safe stopping distances have an important place on determining the traffic rules. The maximum speed allowed in urban roads is 50 km/h. A vehicle travelling with this speed with the conditions given on the table, can stop at approximately l_f=16,42 m when it brakes. When the urban speed limit is 50% increased, on same conditions, brake distance increases 125% and reaches l_b=36,94 m.

When the urban speed limit is exceeded 80% or in other words suburban speed limit of 90 km/h is reached, brake distance increases approximately 225% and reaches l_b=53,19 m.

When the urban speed limit is exceeded 140% or in other words highway speed limit of 120 km/h is reached, brake distance increases approximately 475% and reaches l_b=94,56 m.

As the speed limits determined by the law are being exceeded, brake distances differ a lot. In urban traffic, other vehicles or pedestrians have more interaction with each other so speed limits which obligate short stopping distances are allowed. As highways are high-standard roads, higher speeds are allowed to provide a level of service. In this condition, the brake distances increase, so access control is applied in order to provide safety.

The change of the brake distance for a straigth highway section for various pavement types and conditions are given in the next table. The effect of pavement condition to the brake distance on legal speed limits (50, 90, 120 km/h) are remarkable.

The change of brake distance on a straight road section according to the friction coefficient and the velocity (m)

V(km/h)	*f=0,10*	*f=0,15*	*f=0,20*	*f=0,25*	*f=0,30*	*f=0,35*	*f=0,40*	*f=0,45*	*f=0,50*	*f=0,55*	*f=0,60*
10	3,94	2,63	1,97	1,58	1,31	1,13	0,99	0,88	0,79	0,72	0,66
15	8,87	5,91	4,43	3,55	2,96	2,53	2,22	1,97	1,77	1,61	1,48
20	15,76	10,51	7,88	6,30	5,25	4,50	3,94	3,50	3,15	2,87	2,63
25	24,63	16,42	12,31	9,85	8,21	7,04	6,16	5,47	4,93	4,48	4,10
30	35,46	23,64	17,73	14,18	11,82	10,13	8,87	7,88	7,09	6,45	5,91
35	48,27	32,18	24,13	19,31	16,09	13,79	12,07	10,73	9,65	8,78	8,04
40	63,04	42,03	31,52	25,22	21,01	18,01	15,76	14,01	12,61	11,46	10,51
45	79,79	53,19	39,89	31,91	26,60	22,80	19,95	17,73	15,96	14,51	13,30
50	98,50	65,67	49,25	39,40	32,83	28,14	24,63	21,89	19,70	17,91	16,42
55	119,19	79,46	59,59	47,67	39,73	34,05	29,80	26,49	23,84	21,67	19,86
60	141,84	94,56	70,92	56,74	47,28	40,53	35,46	31,52	28,37	25,79	23,64
65	166,47	110,98	83,23	66,59	55,49	47,56	41,62	36,99	33,29	30,27	27,74
70	193,06	128,71	96,53	77,22	64,35	55,16	48,27	42,90	38,61	35,10	32,18
75	221,63	147,75	110,81	88,65	73,88	63,32	55,41	49,25	44,33	40,30	36,94
80	252,16	168,11	126,08	100,86	84,05	72,05	63,04	56,04	50,43	45,85	42,03
85	284,67	189,78	142,33	113,87	94,89	81,33	71,17	63,26	56,93	51,76	47,44
90	319,14	212,76	159,57	127,66	106,38	91,18	79,79	70,92	63,83	58,03	53,19
95	355,59	237,06	177,79	142,23	118,53	101,60	88,90	79,02	71,12	64,65	59,26
100	394,00	262,67	197,00	157,60	131,33	112,57	98,50	87,56	78,80	71,64	65,67
105	434,39	289,59	217,19	173,75	144,80	124,11	108,60	96,53	86,88	78,98	72,40
110	476,74	317,83	238,37	190,70	158,91	136,21	119,19	105,94	95,35	86,68	79,46
115	521,07	347,38	260,53	208,43	173,69	148,88	130,27	115,79	104,21	94,74	86,84
120	567,36	378,24	283,68	226,94	189,12	162,10	141,84	126,08	113,47	103,16	94,56
125	615,63	410,42	307,81	246,25	205,21	175,89	153,91	136,81	123,13	111,93	102,60
130	665,86	443,91	332,93	266,34	221,95	190,25	166,47	147,97	133,17	121,07	110,98

Pavement condition

f=0,10 : Snowy, wet, unchained
f=0,15 : Basalt small parquet, wet, slightly muddy; icy, wet, chained or dry unchained
f=0,20 : Snowy, wet, chained
f=0,25 : Asphalt, wet, slightly muddy; basalt small parquet, wet, clean; granite small parquet, wet, slightly muddy
f=0,30 : Parquet, wet, slightly muddy; wooden parquet, wet, slightly muddy
f=0,35 : Asphalt, wet, clean; granite small parquet, wet, clean; icy, dry, chained
f=0,40 : Concrete, wet, slightly muddy; rolled ballast, wet, slightly muddy, wooden parquet, wet clean
f=0,45 : Basalt small parquet, dry; snow, dry, unchained
f=0,50 : Granite small parquet, dry; tarmac, dry; crushed ballast, wet, clean
f=0,55 : Concrete, wet, clean; wooden parquet, dry; snow, dry chained
f=0,60 : Asphalt, dry; parquet, dry

The change of brake distance on %5 descending straight road section with respect to the friction coefficient and the velocity (m)

V(km/h)	*f=0,10*	*f=0,15*	*f=0,20*	*f=0,25*	*f=0,30*	*f=0,35*	*f=0,40*	*f=0,45*	*f=0,50*	*f=0,55*	*f=0,60*
10	7,88	3,94	2,63	1,97	1,58	1,31	1,13	0,99	0,88	0,79	0,72
15	17,73	8,87	5,91	4,43	3,55	2,96	2,53	2,22	1,97	1,77	1,61
20	31,52	15,76	10,51	7,88	6,30	5,25	4,50	3,94	3,50	3,15	2,87
25	49,25	24,63	16,42	12,31	9,85	8,21	7,04	6,16	5,47	4,93	4,48
30	70,92	35,46	23,64	17,73	14,18	11,82	10,13	8,87	7,88	7,09	6,45
35	96,53	48,27	32,18	24,13	19,31	16,09	13,79	12,07	10,73	9,65	8,78
40	126,08	63,04	42,03	31,52	25,22	21,01	18,01	15,76	14,01	12,61	11,46
45	159,57	79,79	53,19	39,89	31,91	26,60	22,80	19,95	17,73	15,96	14,51
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60	283,68	141,84	94,56	70,92	56,74	47,28	40,53	35,46	31,52	28,37	25,79
65	332,93	166,47	110,98	83,23	66,59	55,49	47,56	41,62	36,99	33,29	30,27
70	386,12	193,06	128,71	96,53	77,22	64,35	55,16	48,27	42,90	38,61	35,10
75	443,25	221,63	147,75	110,81	88,65	73,88	63,32	55,41	49,25	44,33	40,30
80	504,32	252,16	168,11	126,08	100,86	84,05	72,05	63,04	56,04	50,43	45,85
85	569,33	284,67	189,78	142,33	113,87	94,89	81,33	71,17	63,26	56,93	51,76
90	638,28	319,14	212,76	159,57	127,66	106,38	91,18	79,79	70,92	63,83	58,03
95	711,17	355,59	237,06	177,79	142,23	118,53	101,60	88,90	79,02	71,12	64,65
100	788,00	394,00	262,67	197,00	157,60	131,33	112,57	98,50	87,56	78,80	71,64
105	868,77	434,39	289,59	217,19	173,75	144,80	124,11	108,60	96,53	86,88	78,98
110	953,48	476,74	317,83	238,37	190,70	158,91	136,21	119,19	105,94	95,35	86,68
115	1042,13	521,07	347,38	260,53	208,43	173,69	148,88	130,27	115,79	104,21	94,74
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125	1231,25	615,63	410,42	307,81	246,25	205,21	175,89	153,91	136,81	123,13	111,93
130	1331,72	665,86	443,91	332,93	266,34	221,95	190,25	166,47	147,97	133,17	121,07

Pavement condition

f=0,10 : Snowy, wet, unchained
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f=0,35 : Asphalt, wet, clean; granite small parquet, wet, clean; icy, dry, chained
f=0,40 : Concrete, wet, slightly muddy; rolled ballast, wet, slightly muddy, wooden parquet, wet clean
f=0,45 : Basalt small parquet, dry; snow, dry, unchained
f=0,50 : Granite small parquet, dry; tarmac, dry; crushed ballast, wet, clean
f=0,55 : Concrete, wet, clean; wooden parquet, dry; snow, dry chained
f=0,60 : Asphalt, dry; parquet, dry

Brake time

The time that passes between the moment the driver pushes down the brake pedal and the moment the vehicle stops is named the brake time. The negative acceleration value just after the brake pedal is pushed can be formulated as:

$$a = \frac{v - v_s}{tf}$$

v : The velocity of the vehicle when the brake is applied (m/sec)

v_s : The velocity of the vehicle when it stops (m/sec)

t_f : Brake time (sec)

As the speed of the vehicle is zero when it stops,

$$t_f = \frac{v}{\gamma}$$

However, practically, the deceleration is not on the desired level at the beginning of the braking and reaches the maximum value in a short period of time. The maximum value of the deceleration can be calculated as:

$$\gamma_{max} = f.g$$

If *f,* the sliding friction coefficient; g, the gravitational acceleration are written in the formula,

$$t_f = \frac{v}{f.g}$$

is obtained. If the value of the gravitational acceleration g=9,81 m/sec^2 is written and the velocity is transformed to km/h, the formula takes the form of:

$$t_f = \frac{V}{35{,}3.f}$$

The time past during brake safety distance or in other words, total stopping time for a vehicle moving with a constant speed is the sum of the perception-reaction time and brake time. The calculation of these times are especially important on analyzing the traffic accidents.

3.3.2 Passing-sight distance

On undivided two-way roads, vehicles with a higher speed pass the vehicles with a lower speed with an overtaking maneuver. If the overtaking possibility is unlikely, the higher-speed vehicles continue their movement behind the other vehicle by leaving a distance between them. Especially on ascending sections, if the road is designed without taking the overtaking into account, queing occurs and the capacity of the road decreases.

The drivers driving in a continuous convoy get driving safety. The distance needed for a driver to safely overtake a vehicle moving in front with a lower velocity is named as the passing distance. The passing distance calculated with respect to a specific project velocity corresponds to the minimum sight distance. For this reason, the distance covered in the overtaking maneuver is called the passing-sight distance.

A driver driving with a specific velocity on an undivided road wants to overtake a vehicle moving with a lower velocity in front, makes the overtake manouevre in two different conditions as whether another vehicle is approaching from the opposite direction or not.

Also the overtake manouevre can be made witout increasing the velocity, in other words, without acceleration, or with increasing the velocity, in other words with acceleration according to the road conditions.

3.3.2.1 Vehicle following distance

Vehicles moving as a group on a highway should keep a distance between them in order to provide safety. The distance between the consecutive vehicles is named as the vehicle following distance.

Theoretically, the distance a vehicle should leave between the frontal vehicle while following it is equal to the distance covered between the driver recognizing the brake light of the frontal vehicle and applying the brake, or in other words, the perception-reaction time. However, drivers do not apply a full brake in these conditions and have a short period of hesitation. For this reason, the distance that should be left increases. Experimental studies show that the vehicle following distance can be calculated in meters as:

$$d = a + b.V$$

In this formula;

a = Approximately the vehicle length is about 5~8 m,

b = A constant coefficient determined with respect to the perception-reaction time, which is about 0,20~0,30.

As the perception-reaction times differ in different countries, the formula also changes. As the higher limits of the coefficients are recommended for our country according to the driver behaviors, the formula takes the form of the following:

$$d = 8 + 0{,}3.V$$

Also, the vehicle speed has to be taken as km/h in this formula.

The passing distances for accelerated and unaccelerated movement and whether a vehicle approaches from the opposite direction or not will be studied in the following section.

3.3.2.2 Overtake maneuver on unaccelerated movement

On this type of an overtake manouevre, overtaking, overtaken and the approaching vehicles are accepted to be moving with a constant velocity (the overtaking vehicle's velocity is higher than the overtaken) and do not change their velocities during the movement.

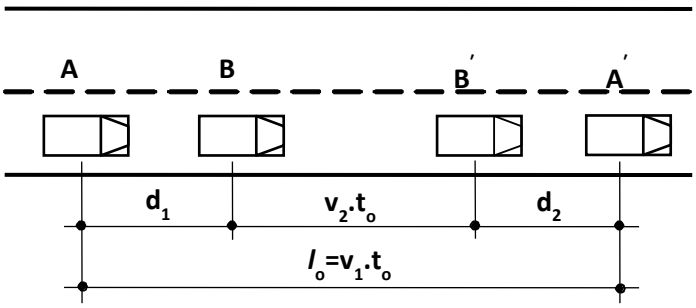
Usually it is valid when there's enough sight distance for the driver to overtake.

Additionally, as the velocity difference with the frontal vehicle is high, even if a vehicle approaches from the opposite direction, acceleration is unnecessary and the driver can overtake in a safe distance.

a) When a vehicle doesn't approach from the opposite direction

The driver of vehicle A, having a velocity of v_1 (m/sec) approaches the vehicle B, having a velocity of v_2 (m/sec) travelling in front for about d_1 meters. If there isn't an obstruction for the

overtake, the driver of vehicle A starts to manouevre. The driver continues to drive on left lane for some time and passes vehicle B. The driver of vehicle A returns to right lane after leaving a distance of d_2 between them and continues driving.



The distance and the time needed for the driver of the vehicle A to finish overtaking maneuver can be formulated as:

$$l_o = v_1.t_o = d_1 + v_2.t_o + d_2$$

If t_o is taken, the following formula is obtained. As the velocities are in m/sec on the formula, they have to be transformed if given in the form of km/h.

$$t_o = \frac{d_1 + d_2}{v_1 - v_2}$$

If t_o is written on the formula, the overtake distance can be written as below. As velocity terms are found on the divider and the divident, there's no need for a transformation.

$$l_o = \frac{(d_1 + d_2)}{(v_1 - v_2)}.v_1$$

t_o : Overtake time (sec)

l_o : Overtake distance (m)

d_1 : The following distance between the overtaking and the overtaken vehicle (before the manouevre) (m)

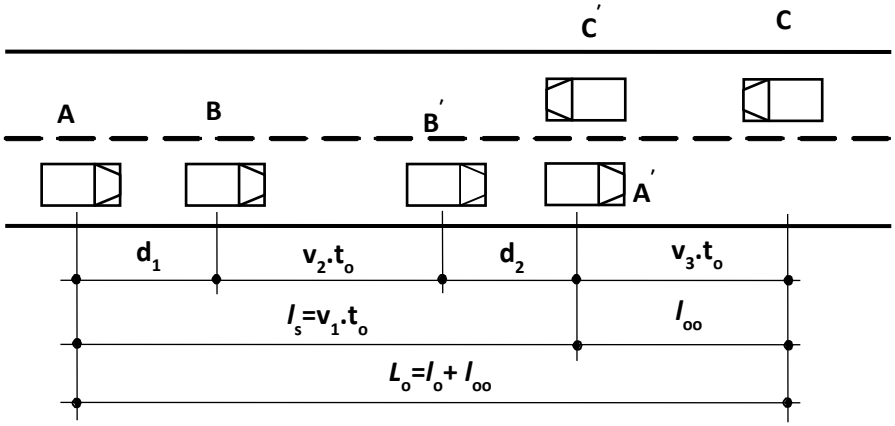
d_2 : The following distance between the overtaken and the overtaking vehicle (after the manouevre) (m)

v_1 : Velocity of the overtaking vehicle (m/sec or km/h)

v_2 : Velocity of the overtaken vehicle (m/sec or km/h)

b) When a vehicle approaches from the opposite direction

On this type of an overtake maneuver, for a safe overtake, the distance covered by the vehicle approaching from the opposite direction during the overtaking maneuver has to be added to the overtaking distance.



So the new passing-sight (overtaking) distance can be calculated as:

$$L_o = l_o + l_{oo}$$

If the formula is written in a more detailed way,

$$L_o = v_1 \cdot t_o + v_3 \cdot t_o = v_1 \cdot \frac{(d_1 + d_2)}{(v_1 - v_2)} + v_3 \cdot \frac{(d_1 + d_2)}{(v_1 - v_2)}$$

If the overtake times are taken out of paranthesis, the formula takes the form of:

$$L_o = \frac{(d_1 + d_2)}{(v_1 - v_2)} . (v_1 + v_3)$$

L_o : Overtake distance when a vehicle approaches from the opposite direction (m)

l_o : The overtaking distance of vehicle A (m)

l_{oo} : The distance covered by the vehicle approaching from the opposite direction during the overtake manouevre (m)

v_1 : The velocity of the overtaking vehicle (m/sec or km/h)

v_2 : The velocity of the overtaken vehicle (m/sec or km/h)

v_3 : The velocity of the vehicle approaching from the opposite direction (m/sec or km/h)

d_1 : The following distance between the overtaking and the overtaken vehicle (m)

d_2 : The following distance between the overtaken and the overtaking vehicle (m)

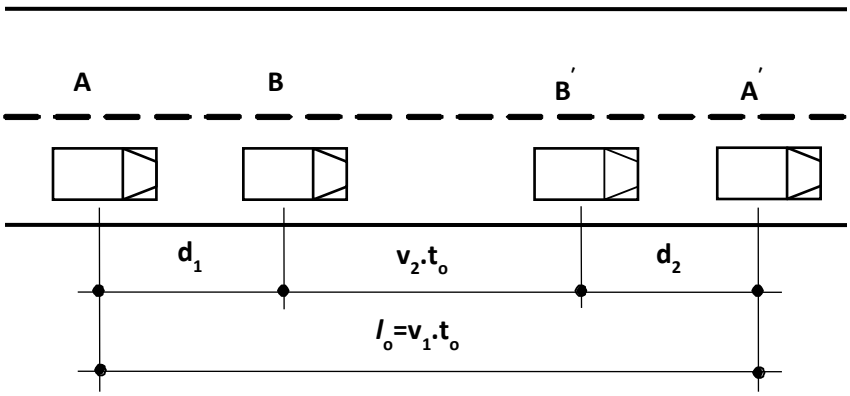
When the highway is being designed or when the traffic signs are being placed, the project velocity has to be used as v_3 because the most inappropriate condition would be the vehicle approaching with the project velocity from the opposite direction. If lower velocities are taken, overtake distance decreases and a dangerous situation may occur.

3.3.2.3 Overtake manouevre on accelerated movement

On this type of an overtake manouevre, overtaking vehicle increases its velocity to overtake the vehicle in front, so the acceleration of the overtaking vehicle is also taken into account.

a) When a vehicle doesn't approach from the opposite direction

The driver of vehicle A, having a velocity of v_1 (m/sec) approaches the vehicle B, having a velocity of v_2 (m/sec) in front for d_1 meters. Then the driver of the vehicle A continues following vehicle B with the velocity of v_2 for some time. If there isn't an obstruction for the overtake, the driver of vehicle A starts to manouevre, increasing its velocity from v_2 to v_1.



There's a short time between the instant the overtaking possibility occurs and the instant the driver recognizes it and starts overtaking maneuver. This time is approximately the same as the perception-reaction time (t_r). On the other hand, the driver is assumed to increase the velocity of the vehicle with an acceleration of a (m/sec^2). As the velocities of the following vehicles are the same before and after the overtaking maneuver, $d=d_1=d_2$ can be written. As vehicle following distances are written as a single d term, the passing-sight distance for accelerated motion can be formulated as:

$$l_{oa} = v_2 \cdot t_2 + 2d = v_2 \cdot t_0 + \frac{1}{2} \cdot a \cdot (t_o - t_r)^2$$

The multiplications having v_2 term on each side of the inequality, is equal to the distance covered by the overtaken (B) vehicle during the overtaking manouevre.

For this reason, $t_o = t_2$ and using $t_{oa} = t_o$, it can be simplified as:

$$2d = \frac{1}{2} \cdot a \cdot (t_{oa} - t_r)^2$$

In conclusion, passing (overtaking) time t_{oa} on accelerated motion can be written as:

$$t_{oa} = t_r + 2.\sqrt{\frac{d}{a}}$$

If the accelerated overtaking time is written on the overtaking distance formula,

$$l_{oa} = v_2 . \ t_r + 2.\sqrt{\frac{d}{a}} \ + 2d$$

l_{oa} : The accelerated overtaking distance of vehicle A (m)

t_r : Perception-reaction time (sec)

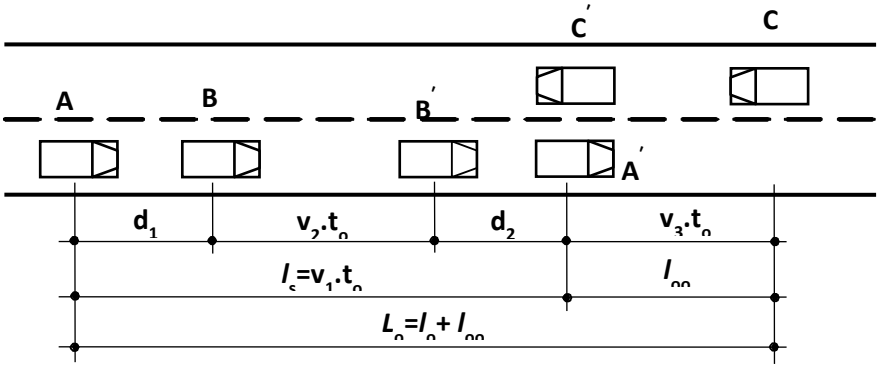
a : The acceleration of the overtaking vehicle (m/sec^2)

v_2 : The volacity of the overtaken vehicle (m/sec)

d : Vehicle following distance (m)

b) When a vehicle approaches from the opposite direction

On this type of an overtake maneuver, as stated before, the distance covered by the vehicle approaching from the opposite direction during the overtaking maneuver has to be added to the accelerated overtaking distance.



The safe accelerated overtake distance can be formulated as:

$$La = l_{oa} + l_{oo}$$

If the formula is extended,

$$L_{oa} = v_2.t_{oa} + 2d + v_3.t_{oa}$$

After the necessary simplifications the formula becomes:

$$L_{oa} = t_r + 2.\sqrt{\frac{d}{a}} .(v_2 + v_3) + 2d$$

L_{oa} : Accelerated overtaking distance when a vehicle approaches from the opposite direction (m)

l_{oo} : The distance covered by the vehicle approaching from the opposite direction during the overtaking manouevre (m)

v_3 : The velocity of the vehicle approaching from the opposite direction (m/sec)